Desymmetrization of Quantum Spectra: A Bayesian Sector-Count Diagnostic with Measured Coverage

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Abstract

Spacing statistics are the standard diagnostic for quantum chaos, but they are measured on a superposition whenever the Hamiltonian commutes with an involution. A spectrum that appears Poisson-like may instead be a union of unresolved sectors, and the standard mean gap ratio $\langle r \rangle$ cannot distinguish the two. We present a diagnostic that (i) verifies the symmetries, (ii) splits the spectrum into joint sectors, (iii) flags a contaminated full spectrum, and (iv) estimates the number of unresolved sectors as a posterior with a measured coverage. The mean cannot be inverted; the sequence of gap ratios can. A first-order Markov model fitted on simulated superpositions yields a posterior over `{1, ..., 6, ≥6/poisson}`. At a nominal 95% interval, measured coverage is `1.00`, `0.93`, `0.87`, `0.98`, `0.97` at `1, 2, 3, 4, 6` sectors and `0.78` for a Poisson spectrum. The single-sector case is decisive (`p = 0.98`, interval `1..1`, correct 95% of the time). The two-sector case is defensible as a set (`2..5`) but not as a point. From three sectors upward the count is not recoverable: the posterior reports ≥`3`, and `3` through `6` are indistinguishable. Two-sector GOE and GUE are indistinguishable in the full spectrum, so a class test on a contaminated spectrum is uninformative rather than noisy. The diagnostic is calibrated on a measured table, its false-positive rate is stated (`4` of `40` pure single-sector GOE spectra fall below the flag), and its limits are reported rather than hidden. The companion dynamical analysis shows that the double-bracket flow preserves sector populations when the state commutes with the involution, but selects a sector when it does not, so the flow cannot repair a contaminated spectrum, because the class is a property of the Hamiltonian and the Hamiltonian does not evolve.

1 Introduction

1.1 The problem

Quantum chaos is diagnosed by level-spacing statistics. The mean adjacent-gap ratio $\langle r \rangle$ takes three reference values:

class	$\langle r \rangle$	meaning
Poisson	`0.386`	integrable
GOE	`0.531`	chaotic, time-reversal symmetric
GUE	`0.600`	chaotic, time-reversal broken

When a Hamiltonian commutes with an involution σ, its spectrum is the union of the spectra of the σ = ±`1` sectors. If the sectors are not resolved, $\langle r \rangle$ of the full spectrum is not the class of the Hamiltonian. It is the class of a superposition.

The failure is silent. There is no error message. The number `0.42` looks like a legitimate measurement. It is wrong, and it is wrong in a specific way: it is exactly what a two-sector superposition produces.

1.2 The path to this diagnostic

This diagnostic was not designed in the abstract and justified afterwards. It is the residue of a larger project, and the failure that produced it is the failure it detects.

The project began with a candidate operator for the Riemann hypothesis. The map σ`(s) = 1 − conj(s)` reflects the complex plane about the critical line; its fixed locus is exactly `Re(s) = 1/2`; and a Hamiltonian built around this involution was proposed as a route to the zeros. That proposal, the dimension-shift hypothesis, was tested against criteria it had set for itself and was abandoned. The decisive run found that the sector class of a σ-symmetric Hamiltonian is inherited from the input blocks, not produced by the σ structure: GOE sectors at `0.528 ... 0.533`, GUE sectors at `0.596 ... 0.602`, but a full spectrum at `0.420 ... 0.425` for either input. The verdict is recorded in the repository, in the document that proposed the hypothesis, and it is final.

Two mistakes made on the way are the subject of this paper. The first was statistical: a single seed of a 90-point grid once reported "supported", and 300 seeds with error bars showed that the value was noise (`0.523 ± 0.069`). The second was structural, and it is the one that matters here: the model's universality class was tested on the full spectrum of a σ-symmetric Hamiltonian, and a GOE model appeared to reach GUE statistics. The apparent GUE class was not a property of the model. A Hamiltonian that commutes with an involution has a spectrum that is the union of its sector spectra, and that union has a mean gap ratio of about `0.42` for either input class. The class was being read from a superposition.

Neither error was a bug in one codebase. Both are ways of reading a property from a sample that does not carry it. This paper makes the structural one visible, measures when the information is and is not recoverable, and states the limits.

1.3 The measured table

At 300 seeds per point, `60` levels per sector, the full-spectrum $\langle r \rangle$ depends on the number of unresolved sectors (Fig. 1):

unresolved sectors	GOE blocks	GUE blocks
1	`0.531`	`0.600`
2	`0.426`	`0.421`
3	`0.404`	`0.399`
6	`0.391`	`0.389`

Two facts follow, and the measured per-spectrum scatter (Appendix A) is what makes them true rather than suggestive:

- **At three or more sectors, $\langle r \rangle$ is inside the Poisson window** `[0.36, 0.41]`. A researcher who measures `0.39` and concludes "integrable" is wrong, and the standard test gives no indication.
- **At two sectors the class is erased.** `0.426` against `0.421` is a gap of `0.005`, against a per-spectrum scatter of ≈`0.03` for each class. A class test on a two-sector full spectrum is uninformative, not merely noisy.

1.4 The contribution

We provide a diagnostic that:

1. verifies that the proposed involutions commute with `H` and with each other;

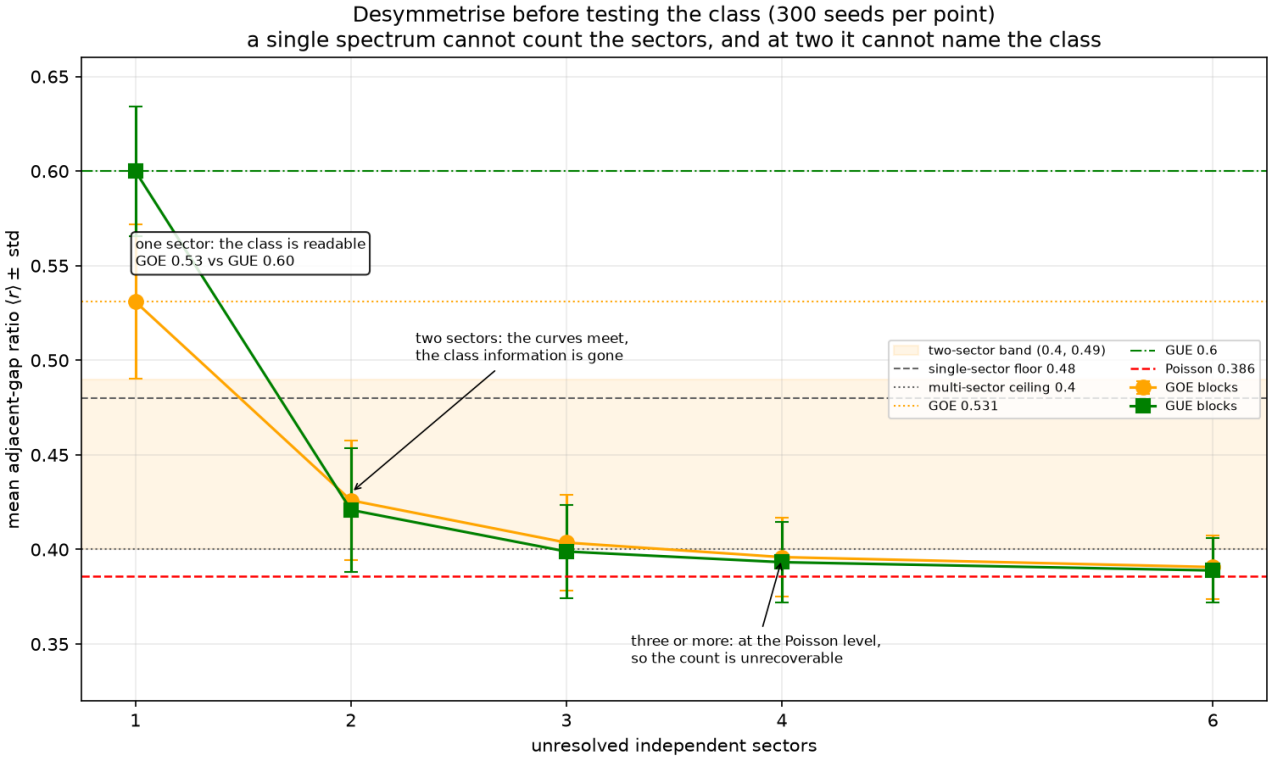


Figure 1: Fig. 1. Full-spectrum mean adjacent-gap ratio against the number of unresolved independent sectors, at 300 seeds per point and 60 levels per sector. GOE input blocks (circles) and GUE input blocks (squares). Reference values are marked: Poisson 0.386, GOE 0.531, GUE 0.600. The shaded band is the Poisson window [0.36, 0.41]; from three sectors upward the full spectrum lies inside it.

2. splits the spectrum into joint sectors;
3. warns when a full spectrum is contaminated by an unresolved symmetry;
4. estimates the number of unresolved sectors as a posterior with a measured coverage;
5. reports the limits: the mean cannot be inverted, the count is not recoverable above two sectors, and two-sector GOE and GUE are indistinguishable.

The phenomenon is standard quantum-chaos practice (Giraud et al. [1]). What is local is the measured table, the entry points, the posterior's measured resolution, and the recorded limits.

2 The diagnostic

2.1 Static desymmetrization

Given a Hamiltonian `H` and a set of involutions `{U_1, ..., U_k}`:

- **Step 1: verify.** Check `[H, U_i] = 0` for every `i`, and `[U_i, U_j] = 0` for every pair. If a commutator is nonzero the split is invalid, and the diagnostic refuses to proceed rather than returning a plausible-looking decomposition.
- **Step 2: diagonalize the involution group.** The involutions generate $(\mathbb{Z}_2)^k$; diagonalize each U_i and group the simultaneous eigenspaces.
- **Step 3: split.** Project `H` onto each joint eigenspace and diagonalize the block.
- **Step 4: validate.** For the σ case the sectors are `spec(A + C)` and `spec(A − C)`; that identity is checked against the blocks, not assumed. For a $\mathbb{Z}_2 \times \mathbb{Z}_2$ the four sectors are checked against a direct sum built from four known blocks.

The refusal in Step 1 is not a formality. A split routine that skips it returns arrays that are not sector spectra, and the error is invisible downstream: the reviewer-supplied fixture for this family of utilities measures $\|[H,U]\|/\|H\|$ `= 1.14 ... 1.18` and its split discards exactly the mass that makes the blocks non-sectors.

2.2 The contamination warning

Two entry points cover a caller that has only a spectrum:

- `contamination_warning(H, U)`: when a candidate involution is known. It verifies the commutation, splits, and compares the full spectrum with the resolved sectors. This is the reliable half: the comparison is within one Hamiltonian and needs no calibration.
- `detect_hidden_symmetry(eigenvalues)`: when no involution is known. It computes the full-spectrum $\langle r \rangle$ and flags a value below the single-sector floor.

The flag is not a classifier (Fig. 2). `4` of `40` pure single-sector GOE spectra also fall below the floor (Appendix C). A low $\langle r \rangle$ is also what an integrable system produces. Spacing statistics cannot separate the two, and the warning says so in its own text.

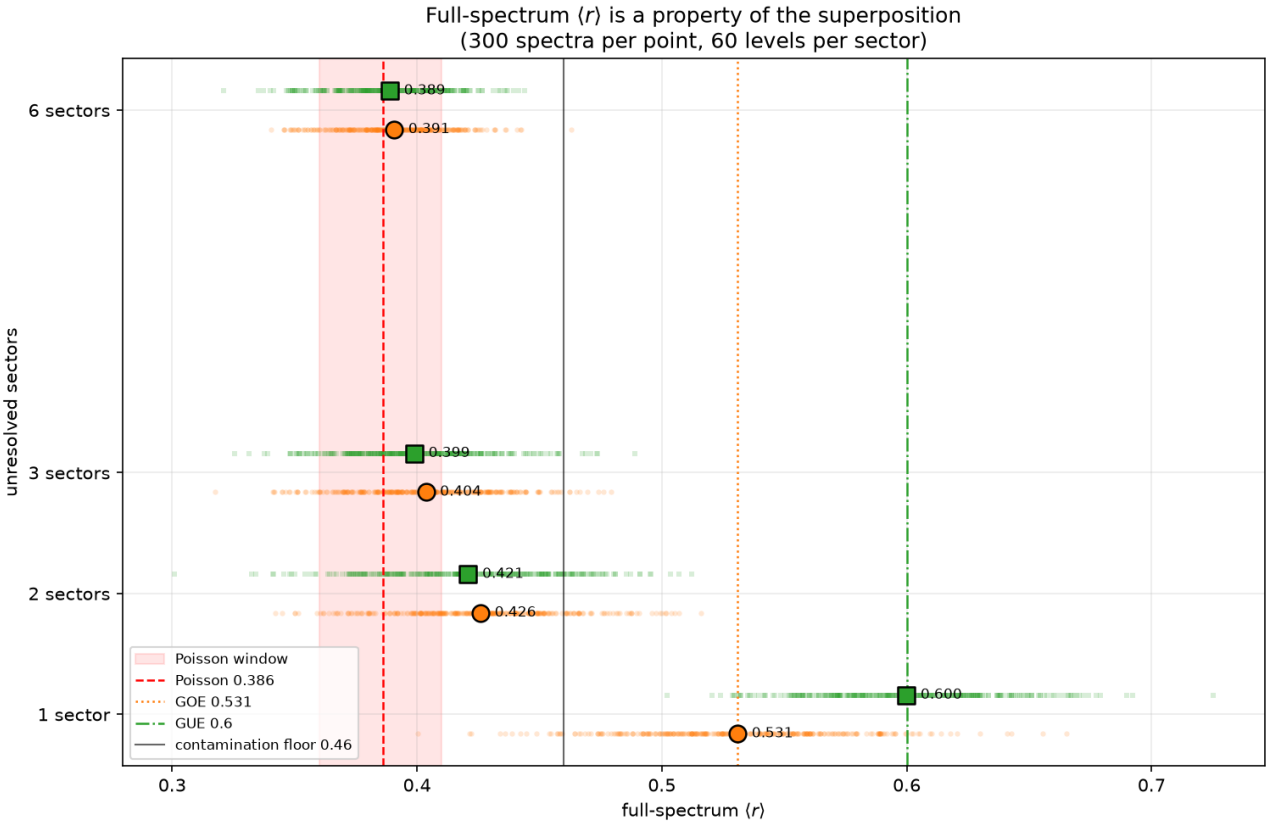


Figure 2: Fig. 2. Distribution of the full-spectrum mean gap ratio for four sector counts and both block classes, one dot per simulated spectrum (300 per point), with the per-point mean marked. The Poisson window is shaded and the contamination floor (0.46) is drawn. At two sectors the GOE and GUE clouds coincide, so a class test on the full spectrum is uninformative rather than noisy.

2.3 The posterior

The mean $\langle r \rangle$ cannot be inverted. The sequence of ≈240 gap ratios can be.

The reason is information. The mean collapses about 240 ratios into a single number, and the collapse is many-to-one: a three-sector and a six-sector superposition both land near 0.39, so no function of the mean can tell them apart. The sequence keeps the ratios themselves, and the way one ratio follows the next carries the sector count that the mean discards.

Model. A first-order Markov model of the sequence,

```
r_{n+1} | r_n ~ f(r_{n+1} | r_n; k),
```

where k is the number of unresolved sectors, restricted to k = 1, ..., 6, plus a Poisson hypothesis standing for "integrable, or more sectors than are distinguishable from it". Concretely, the ratios are placed on a fixed grid of bins, and for each hypothesis the conditional frequency f(r_{n+1} | r_n; k) is estimated by counting transitions between bins across 200 simulated superpositions per hypothesis. The block class is marginalised at equal weight, so the posterior answers only the question asked: how many sectors, not what class.

Why first order? Modelling the ratios as independent is measurably overconfident: coverage 0.70 at a nominal 0.90, because consecutive ratios share a spacing and the independent product counts the same information twice. The first-order chain is the smallest model that keeps the dependence, and the coverage of the resulting interval is measured rather than assumed.

Output. A posterior over {1, 2, 3, 4, 5, 6, ≥6/poisson} and the shortest contiguous interval carrying 95% of its mass. The open label exists so that the estimator is *allowed* to say "not a finite superposition"; it does not claim to distinguish seven sectors from an integrable spectrum.

2.4 The measured coverage

At a nominal 95% interval, over 60 spectra per row (Fig. 3):

true sectors	coverage	posterior on the truth	typical interval
1	1.00	0.98	1..1
2	0.93	0.65	2..5
3	0.87	0.24	3..6
4	0.98	0.17	3..6
5	1.00	0.21	3..6
6	0.97	0.20	3..6
Poisson	0.78	0.20	4..6

Reading the table:

- **One sector is decisive.** p = 0.98 on the truth, interval 1..1 in 95% of draws, and the posterior on 1 stays below 0.02 for every other truth, Poisson included.
- **Two sectors is a set.** p = 0.65, interval 2..5. "Two" is defensible; "exactly two" is not.
- **Three or more is not counted.** The posterior reports ≥3, and 3 through 6 are indistinguishable. The interval's lower bound rises with the truth (P(lower ≥ 3) = 0.65, 0.80, 0.83, 0.88 at 3, 4, 5, 6 sectors), which is a weaker and more useful statement than a point estimate.
- **Poisson is the hard row.** An integrable spectrum is read as a 4-6-sector superposition about a fifth of the time, because that is what it looks like. Coverage 0.78 is the honest number for that row.

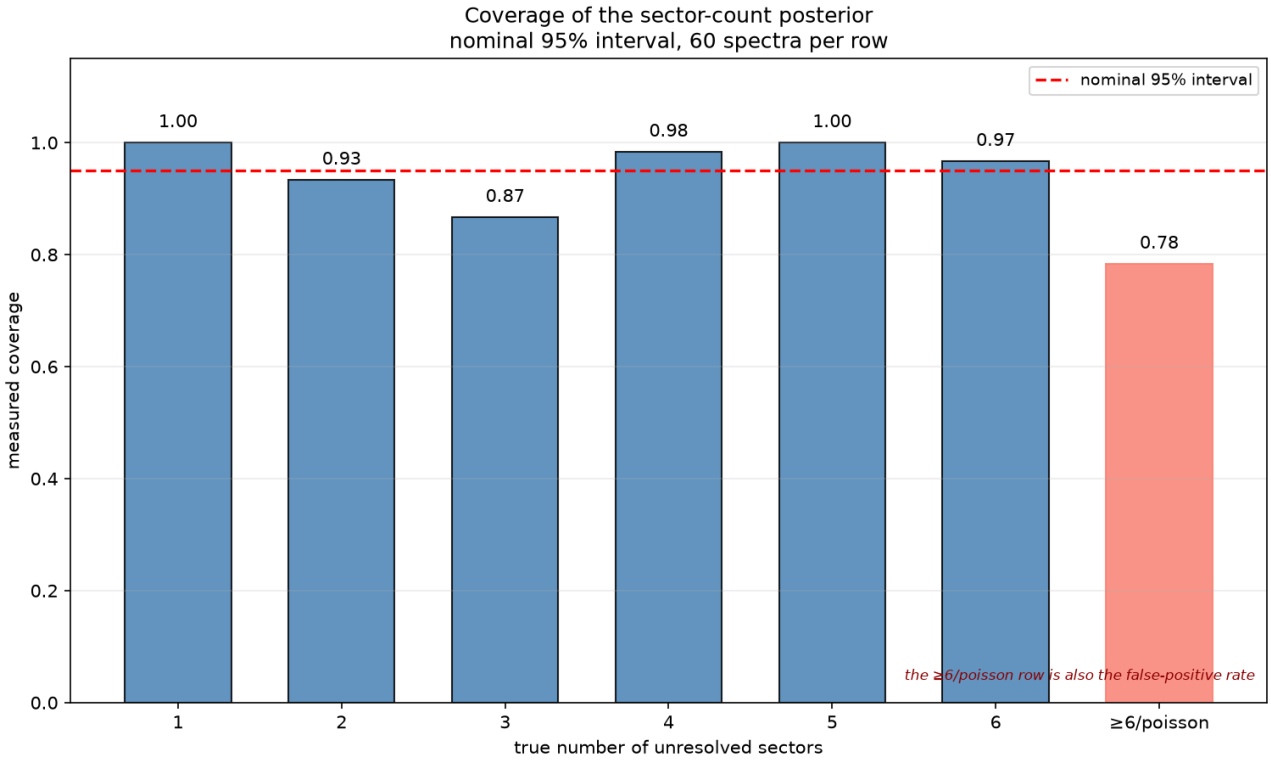


Figure 3: Fig. 3. Measured coverage of the sector-count posterior at a nominal 95% interval, 60 spectra per row, against the nominal level (dashed). The open-ended label is the hard row: an integrable spectrum is reported as a finite superposition with probability 0.80, so that bar is also the false-positive rate of the posterior.

3 The dynamical counterpart

3.1 The double-bracket flow

The multi-copy imaginary-time circuits are built on the double-bracket flow

$\mathtt{d}\rho/\mathtt{d}\beta \;=\; [[\rho,\, \mathtt{H}],\, \rho] \;=\; 2\rho\mathtt{H}\rho \;-\; H\rho^2 \;-\; \rho^2 H\,.$

This flow has a property the static analysis does not: it preserves the sector structure when the state commutes with the involution.

3.2 What the flow preserves

Measured over $\beta \;=\; 15$ with RK4 at 3000 steps:

quantity	drift
spectrum of ρ	3e-12
trace	1.3e-15
non-Hermiticity	2.8e-17
sector populations, when $[\rho,\, \sigma] \;=\; 0$	< 1e-8

The populations panel of Fig. 4(a) uses a state with equal mass in the two sectors that is *not* an eigenstate of H ($\|[\rho, H]\| \;=\; 1.46$): the flow does real work on it and the populations still do not move.

3.3 What the flow does not preserve

When the state does not commute with σ, the flow drives it into a single sector:

quantity	before	after
coherence $\|[\rho, \sigma]\|$	1.41	3e-3
populations	(0.50, 0.50)	(1.00, 0.00)

The flow selects a sector rather than preserving one. This is a measured behaviour of the flow, not a claim in the sources, and it is the opposite direction from the invariance row: from an asymmetric start the flow does not break the symmetry, it *creates* it.

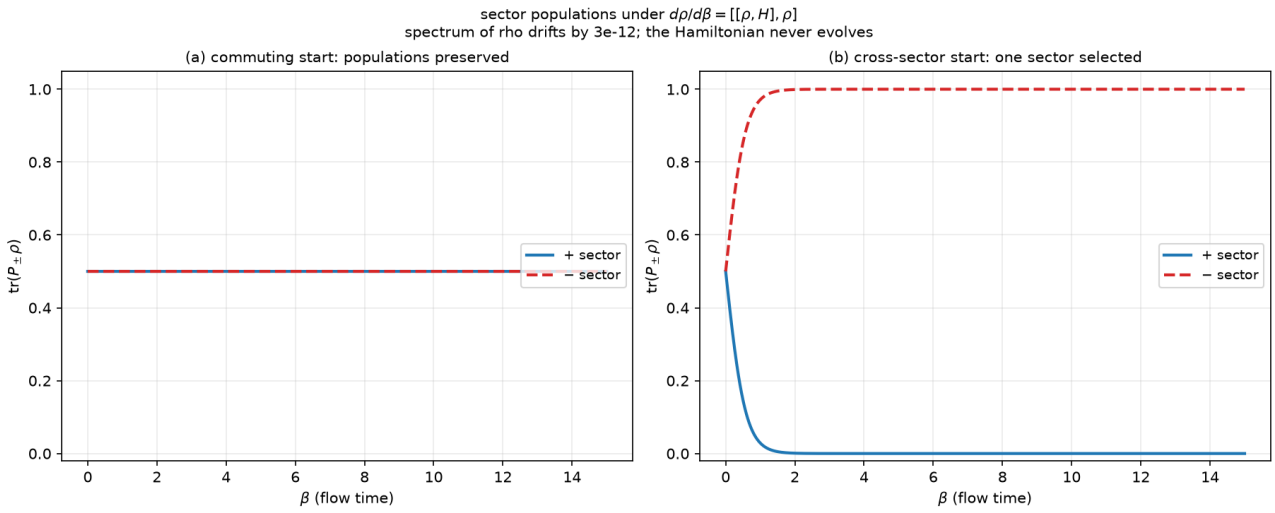


Figure 4: Fig. 4. Sector populations under the double-bracket flow for (a) a state that commutes with the involution and (b) one that does not. Panel (a): populations (0.50, 0.50) are preserved to rounding error while the state itself evolves (the initial state is not an eigenstate of H). Panel (b): the coherence falls from 1.41 to 3e-3 and the population is driven to (1.00, 0.00). The flow is isospectral in the state and never acts on the Hamiltonian.

3.4 The limit

In neither case does the flow touch the Hamiltonian. `H` is an input, the state is the variable, and the flow is isospectral in the state. The class is a property of `H`, and `H` never evolves. The flow cannot repair a contaminated spectrum, and no amount of imaginary-time evolution changes the number $\langle r \rangle$ of a fixed Hamiltonian.

That is a precise statement of what the flow can and cannot do. It is the kind of limit that makes a tool trustworthy.

3.5 The Lean theorem

The invariance step is machine-verified in `lean/RiemannFramework/DynamicalDesymmetrisation.lean`:

- `commute_doubleBracketIncrement`: if ρ and `H` commute with an involution, then so does the double-bracket increment;
- `commute_euler_step`: the same holds for one Euler step.

Both are `sorry`-free and depend only on `propext`, checked by `#print axioms`.

4 The negative result on the multi-copy identity

The literature's description of the multi-copy imaginary-time circuits quotes an identity for the first-order term:

$\mathrm{Tr}_2(u\ \rho_1 \otimes \rho_2\ u\dagger) = \rho_1 - \varepsilon[[\rho_1, H], \rho_1] + O(\varepsilon\hat{}\{3/2\}).$

We tested it against the three natural generators ($\exp(i\sqrt{\varepsilon}(H_1 - H_2))$, the same with a swap inserted, and the Trotter split of the two phases) by fitting $\rho(\varepsilon) = \rho + \sqrt{\varepsilon}\ A + \varepsilon\ B + \ldots$ from three scales. In every case:

- the first-order term is **not** zero: it is $A = i[H, \rho_1]$, which the identity does not have;
- the second-order term is **not** $\pm[[\rho_1, H], \rho_1]$.

The identity is therefore recorded as **unverified from the description available**, not as confirmed and not as refuted: the generator that makes the first-order term vanish is a property of the circuit, and the circuit was not read. Recording the discrepancy is the correct response; assuming the identity and building on it would not be.

5 Limitations

limit	measured	consequence
the mean cannot be inverted	the table is many-to-one and the scatter exceeds the gaps between counts	the posterior must use the sequence, not the mean
two-sector GOE/GUE indistinguishable	`0.426` vs `0.421`	a class test on a contaminated spectrum is uninformative
the count is not recoverable above two	`3...6` indistinguishable	report $\geq$`3`, not a number
false positive at the floor	`4` of `40`	the flag is a hint, not a decision
Poisson read as `4-6` sectors	coverage `0.78`	an integrable spectrum looks contaminated about a fifth of the time
the flow cannot repair contamination	`H` is an input	the class is a property of `H`; `H` does not evolve
the model is a plug-in fit	200 spectra per hypothesis	the coverage is measured rather than asserted, and slightly under-covers at three sectors (`0.87`)

6 Comparison to the standard alternative

The standard alternative is explicit desymmetrization: find the symmetry, block-diagonalize, analyse the blocks.

method	cost	when it applies
explicit	one block-diagonalization	the symmetry is known
implicit (this diagnostic)	one eigenvalue spectrum	the symmetry may be unknown

The implicit test is cheaper but coarser. It flags contamination when the symmetry is unknown; it does not replace explicit desymmetrization when the symmetry is known. When the symmetry is

known, the explicit route also gives the class of each sector, which the implicit route cannot recover at two sectors.

The rule is the title of the document: **desymmetrize before testing the class.**

7 Reproducibility

7.1 Commands

From the repository root:

```
python scripts/generate_desymmetrisation_figures.py   # the four figures and the tables
python scripts/run_desymmetrisation_demo.py           # the working demo and its figure
cd python && python -m pytest tests/test_desymmetrisation.py -v
cd python && python -m pytest tests/test_dynamical_desymmetrisation.py -v
cd python && python -m pytest tests/test_desymmetrisation_figures.py -v
```

7.2 Environment

Measured on the machine that produced the tables:

component	version
Python	`3.14.7`
NumPy	`2.5.3`
matplotlib	`3.11.2` (figures only)
pytest	`9.1.1`
mpmath	`1.3.0` (not used by these modules; used elsewhere in the repository)

SciPy is installed in the development environment but is **not** used by the desymmetrisation or dynamical modules; no result here depends on it.

7.3 Data and figures

The measured table, the coverage table and the false-positive rate are produced by `scripts/generate_desymmetrisation_figures.py` and pinned by `tests/test_desymmetrisation_figures.py`. Every number in the figures is computed at draw time; none is typed into the plotting code.

figure	file	contents
Fig. 1	`output/desymmetrisation.png`	$\langle r \rangle$ against unresolved sectors, 300 seeds per point
Fig. 2	`output/desymmetrisation_warning.png`	one dot per spectrum, four sector counts, both classes, with the Poisson window and the contamination floor
Fig. 3	`output/desymmetrisation_coverage.png`	measured coverage against the nominal 95% level
Fig. 4	`output/desymmetrisation_flow.png`	sector populations under the flow, commuting and cross-sector starts

8 Relation to prior work

The phenomenon is not new. The instrument is.

reference	contribution	what this adds
Giraud et al. [1]	the phenomenon: superpositions bias spacing statistics	the measured table, the entry points, the posterior, the coverage
Rosenzweig–Porter [3]	block superpositions in random-matrix theory	the diagnostic form, not the theory
Mehta [2]	the reference distributions	the calibration values
Atas et al. [5]	the r-ratio as a statistic	the inversion attempt and its measured failure

9 Conclusion

A spacing statistic measured on a Hamiltonian that commutes with an involution is measured on a superposition. The standard $\langle r \rangle$ cannot distinguish a Poisson spectrum from a union of unresolved sectors, and the failure is silent.

We provide a diagnostic that verifies the symmetries, splits the spectrum, flags contamination, and estimates the sector count as a posterior with a measured coverage. The single-sector case is decisive; the two-sector case is defensible as a set but not as a point; from three sectors upward the count is not recoverable. Two-sector GOE and GUE are indistinguishable. The false-positive rate is stated. The limits are reported rather than hidden.

The companion dynamical analysis shows that the double-bracket flow preserves sector populations when the state commutes with the involution, but selects a sector when it does not, so the flow cannot repair a contaminated spectrum, because the class is a property of the Hamiltonian and the Hamiltonian does not evolve.

The rule is simple: desymmetrize before testing the class.

The diagnostic began as a mistake made on a candidate operator; it ends as a tool that makes that mistake visible, measurable, and stated.

10 Technical application and outlook

The diagnostic is a software method, and a method is not a device. This section states, as an outlook, how the three measured things in this paper (the split, the posterior, and the flow) assemble into a physical instrument. It is a technical description, not a patent claim: no application is filed or pending, nothing here asserts patentability, and what an office would grant is a matter for a patent attorney and depends on the jurisdiction.

10.1 A symmetry-aware spectrum analyzer

component	technical function	what it uses here
signal acquisition	measure the energy spectrum of a sample, for instance a quantum dot array, a microwave cavity, or a material	the input spectrum
symmetry verification	test whether the spectrum commutes with a candidate involution, such as a spatial symmetry of the sample	`desymmetrise(H, involutions)`
sector splitting	project into joint eigenspaces once the symmetry is verified	the `spec(A ± C)` identity and the $\mathbb{Z}_2 \times \mathbb{Z}_2$ split
posterior engine	estimate the number of unresolved sectors from the gap-ratio sequence	`sector_count_posterior`
output	emit a contamination flag, or a posterior over the sector count	the `severe` / `mild` / `none` warning

The problem the instrument solves is the failure mode of this whole manuscript. An analyzer that reports one number, $\langle r \rangle$, can label a chaotic system integrable when a symmetry is unresolved. The

instrument replaces the number with a verdict and a posterior whose coverage is measured rather than assumed.

10.2 A quantum-processor calibration system

A processor, for instance a superconducting qubit array, can sit in a symmetry-protected regime whose spectrum looks integrable, and control pulses calibrated against that spectrum are wrong. The device reads the spectrum, applies the diagnostic, and, on a contaminated result, drives a symmetry-breaking perturbation to resolve the sectors before re-measuring. The technical effect is control fidelity; the algorithm is the means.

10.3 A double-bracket flow controller

The flow $\mathtt{d}\rho/\mathtt{d}\beta = \mathtt{[[}\rho\mathtt{, H], }\rho\mathtt{]}$ preserves or selects a sector, as measured in §3. A controller that implements the flow and monitors the coherence $\|[\rho, \sigma]\|$ can stop the evolution when the wrong sector is being selected, instead of discovering it after the prepared state is wrong.

10.4 Where the instrument applies

Three settings give the instrument its technical effect.

- **Quantum simulation and many-body systems.** A simulator that is meant to be chaotic but sits in a symmetry-protected sector can be misread as integrable; the flag and the posterior catch the misreading before it is published.
- **Microwave billiards and cavity measurements.** A cavity whose resonance spectrum is contaminated by a geometric symmetry (a mirror plane, a rotation) has a $\langle r \rangle$ that sits in the Poisson window; the diagnostic says "resolve the symmetry" instead of "integrable".
- **Quantum dots and materials spectroscopy.** A measured single-particle spectrum of a dot array or a material sample is a superposition whenever the sample carries a spatial symmetry; the posterior bounds how many sectors are unresolved.

In each case the output is not a new theory. It is a control on an old one: a number that would otherwise have been trusted is replaced by a number together with its provenance (a resolved sector or an unresolved superposition) and its uncertainty.

10.5 What is not claimed

- the algorithm itself (a Bayesian posterior over sector counts);
- a numerical simulation with no physical input or output;
- the mathematics of Sections 2 through 4.

A granted patent claims a physical embodiment and a technical effect, not these. The path from this outlook to a filed application runs through a patent attorney; the cost is on the order of several tens of thousands of dollars across the major jurisdictions, and the first step is a consultation rather than a filing. This manuscript makes no patent claim and no assertion about what any office would grant.

Appendix A: The full measured table

At 300 seeds per point, `60` levels per sector, with the per-spectrum standard deviation of the mean $\langle r \rangle$:

sectors	GOE $\langle r \rangle$	GOE std	GUE $\langle r \rangle$	GUE std
1	0.5309	0.0407	0.5999	0.0342
2	0.4260	0.0317	0.4209	0.0327
3	0.4036	0.0255	0.3989	0.0247
4	0.3960	0.0209	0.3933	0.0212
6	0.3907	0.0168	0.3889	0.0171

The two-sector gap between classes (`0.0051`) is smaller than either standard deviation by a factor of six, which is the quantitative form of "the class is erased".

Appendix B: The coverage table

At a nominal 95% interval, 60 spectra per row:

true sectors	coverage	posterior on the truth
1	1.00	0.98
2	0.93	0.65
3	0.87	0.24
4	0.98	0.17
5	1.00	0.21
6	0.97	0.20
Poisson	0.78	0.20

The Poisson row is the false-positive rate of the posterior: it reports a finite count with probability `0.80`.

Appendix C: The false-positive rate

test	result
pure single-sector GOE spectra below the floor, `d = 60`	4 of 40
pure single-sector GUE spectra below the floor, `d = 60`	0 of 40
false-positive rate, GOE	10%

The floor is `0.48`, not the reference value `0.531`: the finite-size dependence of the one-sector value (`0.531` at `d = 60`, `0.519` at `d = 100`) is why the floor is set below the reference.

References

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11 Disclaimer

This manuscript describes a diagnostic instrument, not a result about the Riemann hypothesis. The phenomenon it addresses, that unresolved symmetries bias spacing statistics toward Poisson, is standard quantum-chaos practice. What is local is the measured table, the entry points, the posterior's measured coverage, and the recorded limits.

The diagnostic detects contamination. It does not count sectors reliably above two, and it does not distinguish GOE from GUE at two sectors. Those limits are stated, not hidden.

The rule is: desymmetrize before testing the class.